Neexdzii Kwah Restoration Planning

Executive Summary

S E R N B C

SOCIETY FOR ECOSYSTEM RESTORATION
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Prepared for the Wet'suwet'en Treaty Office Society

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Executive Summary

This report presents a governance and prioritization framework for restoration of the Neexdzii Kwah (Upper Bulkley River) watershed, grounded in Wet'suwet'en stewardship values and supported by an integrated suite of open-source spatial analysis tools. The framework is not a static plan — it is a living system designed to evolve as understanding deepens and new data becomes available.

Governance and prioritization. The prioritization framework proposes a four-tier governance structure — Stewardship Council, Technical Working Group, Implementation Partnerships, and Community Roundtable — that places Wet'suwet'en direction at the centre of restoration decision-making. Three diagnostic gates filter candidate sites before investment: certainty of diagnosis, whether active degradation is ongoing, and access and willingness to act. Scoring principles were developed from community workshops where participants ranked restoration parameters. Workshop groups converged on the same four priorities — Cultural Significance, Probability of Success, Watershed Function, and Proximity to High-Value Fish Habitat — though the relative weighting differed across groups. These perspectives are built into the weighting system. The framework acknowledges that habitat restoration alone cannot deliver salmon recovery. Skeena chinook and coho face sustained declines driven by fishing pressure — both marine and in-river — that exceeds what these populations can sustain. Reducing harvest is the necessary precondition; without it, restored habitat produces fish that are removed before they can spawn. Rebuilding habitat capacity now ensures the watershed can support recovery when harvest rates are brought in line with current productivity.

Sub-basin delineation. The watershed was divided into 14 sub-basins to enable prioritization and facilitate communication of issues and opportunities within each area. Natural break points — Bulkley Falls and Buck Falls — define the primary divisions due to their significance as barriers to salmon access. Within these regions, sub-basins were delineated based on groupings of habitat and landscape characteristics at comparable scales, using FWA-referenced spatial hydrology tools via `fresh` and the interactive `breaks` sub-basin delineation tool. Each sub-basin integrates fish habitat modelling, land ownership, reserve lands, cultural sites, and land cover change into a single characterization that supports comparison and investment targeting.

Floodplain modelling and land cover change. The functional floodplain — where the river does geomorphic work including sediment storage, nutrient exchange, riparian recruitment, and large woody debris delivery — was modelled using the Valley Confinement Algorithm implemented in `flooded`. Within the modelled floodplain, 10 m resolution satellite imagery shows approximately 760 ha of tree cover lost across the watershed over the six-year period from 2017 to 2023 — the window for which classified land cover data is available. Tree loss rates vary by sub-basin, from less than 1% to roughly 10% of floodplain area, with what the analysis classifies as agricultural land expanding into cleared areas in most cases. The floodplain extent was modelled conservatively — scoped to coho-accessible streams of 3rd order and larger, with lakes and wetlands included only where they connect to the fish-accessible network.

Historic aerial photographs. Nearly 10,000 provincial aerial photographs (1963–2019) have been downloaded and georeferenced using `fly`. This collection represents the entire provincial dataset where photograph footprints land within the Neexdzi Kwah watershed. It is served as a queryable spatial catalog — available programmatically and from within GIS systems such as QGIS. The interactive `diggs` application allows users to select photos based on centroid and estimated footprint overlap with custom areas of interest (ex. modelled floodplain extents). This 50-year visual record allows direct comparison of historic and current conditions at any location in the watershed — building understanding of how the landscape has changed over time and grounding restoration targets in documented prior condition.

Site assessments. Field reviews at 26 locations over 2024–2025 documented conditions at past restoration sites, historic prescription locations, and newly proposed areas. A recurring theme was riparian and floodplain vegetation removal, channelization of streams and cattle-related degradation at scales far exceeding the footprint of past restoration investments. At sites where past work has been completed, riparian buffer widths were narrower than what current understanding suggests is needed for effective recovery. Road and rail erosion protection consistently lacked vegetated riprap, soft armouring, and effective riparian buffers. UAV flights at the majority of sites provide high-resolution baseline imagery for future monitoring, cataloged via `stac_uav_bc`. These observations feed directly into the prioritization framework and provide the baseline against which future restoration effectiveness will be measured.

Collaborative GIS. A shared QGIS project integrating provincial background layers, field data collection forms, spatialized historic restoration prescriptions, and all analysis outputs serves as the spatial integration layer for this work. Through cloud-based mobile GIS, the full project travels into the field on mobile devices — functioning as both the primary navigation system during site visits and the platform for standardized data collection. The ability to overlay fish habitat modelling, historic prescriptions, floodplain change detection, land tenure, and traditional knowledge in a single environment allows users to understand the spatial context of each location and make informed decisions about restoration priorities. Critically, this is not just an analytical tool — it is the primary platform for field data collection and the spatial integration layer that connects all other project components.

Aquatic health monitoring. Benthic invertebrate sampling at three mainstem sites reveals a clear gradient: stream health improves with distance upstream from Houston. Near town, the community includes a higher proportion of nutrient-tolerant species — midges and net-spinning caddisflies — and the biotic index indicates possible slight organic enrichment, consistent with elevated phosphorus documented in the historical water quality record for this reach. Moving upstream, these give way to sensitive mayflies, stoneflies, and caddisflies that indicate clean conditions. The upstream site below McQuarrie Creek supports reference-quality conditions suitable as a benchmark for future monitoring. Statistical testing confirmed these are genuinely different

communities, not random variation. Compared to previous sampling in 2004 and 2018, the near-Houston community appears to have shifted toward more pollution-tolerant species, though differences in sampling season complicate direct comparison. Full results including approximately 50 years of compiled water quality data are presented in a dedicated report, with data to be uploaded to the federal CABIN database. An expanded monitoring network across the watershed is proposed.

Environmental DNA (eDNA) sampling targeting chinook, coho, sockeye, and bull trout was conducted below Bulkley Falls and approximately 1 km downstream to detect species presence within the reach at the time of collection, with analysis pending from the University of Northern British Columbia.

Tools as deliverables. The spatial analysis tools developed for this work are open-source R packages designed to work at any scale and in any watershed in the province:

- Stream network spatial hydrology — `fresh`
- Interactive sub-basin delineation — `breaks`
- Floodplain delineation — `flooded`
- Land cover change detection — `drift`
- Airphoto footprint estimation and optimal coverage selection — `fly`
- Interactive airphoto explorer — `diggs`
- Provincial LiDAR digital elevation models (DEMs) — `stac_dem_bc`
- UAV imagery catalog — `stac_uav_bc`
- Historic airphoto catalog — `stac_airphoto_bc`

These tools were developed across multiple projects and through ongoing research investment — they are not single-use products tied to any one initiative. The capacity they build — for ourselves and for others — to understand and manage conservation and restoration is as much a deliverable as the analysis results themselves.

This report is a living document, with ongoing revisions and updates tracked in the project changelog. The research and monitoring programmes described here are not just technical exercises — they are acts of territorial stewardship, grounded in the understanding that health, ecosystems, social systems, and watershed management must be governed together (Harris 2011). The United Nations Decade on Ecosystem Restoration (2021–2030) reinforces these principles through international standards of practice that call for addressing root causes of degradation, benefiting the people with the deepest connection to the ecosystem, and integrating governance across scales (FAO et al. 2023). We are humbled by the depth of knowledge that exists for this region, both through traditional knowledge and scientific study. Recovery efforts must go beyond technical interventions and include a commitment to learning from the vast body of knowledge held by the Wet'suwet'en people — including cultural practices, oral tradition (Kungax), Wet'suwet'en Law (Ink Nu'at'en), and the deep connections between families, house groups, and the lands and waters that sustain them (University of British Columbia Library, n.d.; Morin 2016).

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